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GREEK COLONIES, STEPPE NOMADS AND COSSACK TOWNS ON THE LOWER DON AND AZOV COAST

Bosporan Ports, Khazar Fortresses, Golden Horde Azak and Don Cossack and Peasant Settlement

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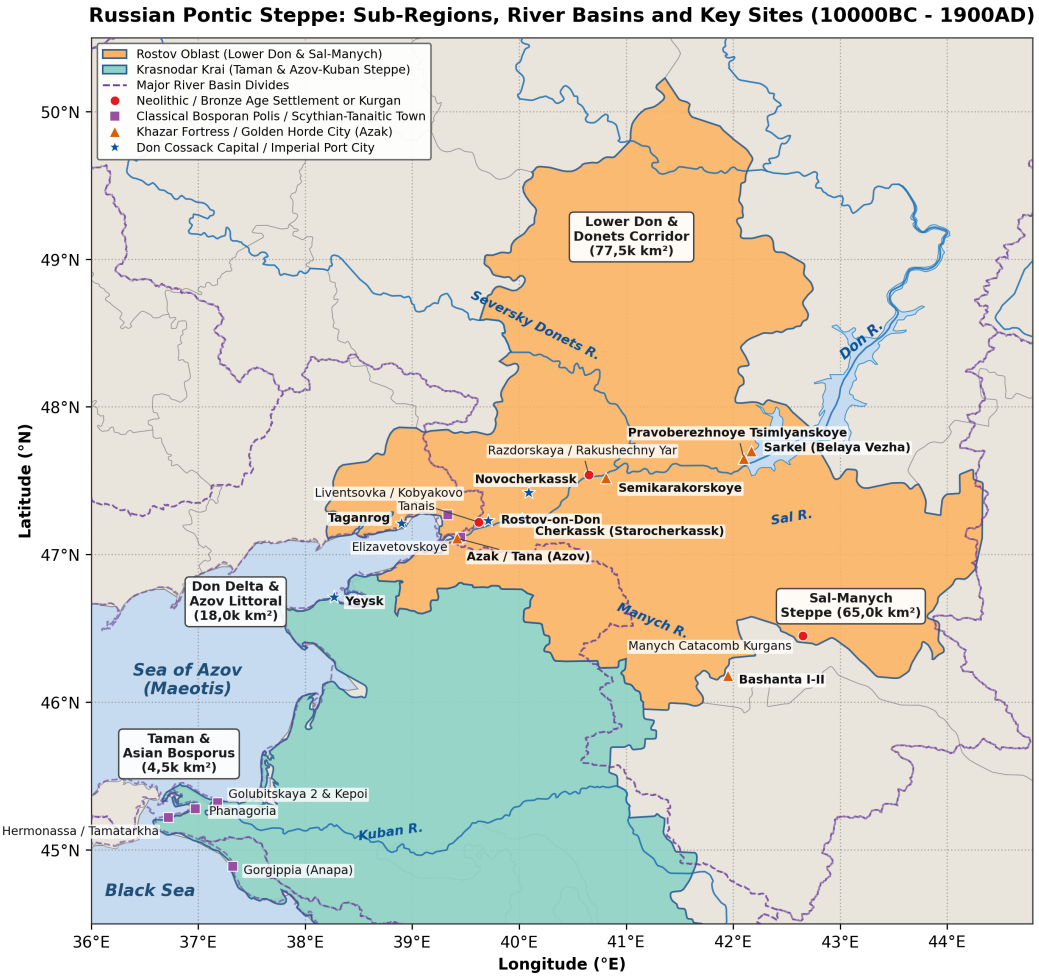


Figure 1. The four regions of the Russian Pontic Steppe (165k km^2), linking the Lower Don Delta, the Taman Peninsula and the Don-Donets valley corridor to the dry inland Sal-Manych steppes.

Abstract

Spanning 165k km^2 between the Sea of Azov, the Lower Don and Seversky Donets valleys, the Taman Peninsula and the Sal-Manych watershed, the Russian Pontic Steppe was for millennia a contested borderland between maritime trading cities and steppe pastoralists. Early riverine fishing villages at Rakushechny Yar supported 2,6k-10,7k inhabitants between 10000BC and 5000BC [1][2], before wheeled

steppe pastoralism in the Yamnaya and Catacomb periods (43k in 2400BC) [3][4] and permanent river-terrace farming under the Late Bronze Age Srubnaya and Kobyakovo cultures lifted the regional total to 80k by 1500BC [5][6]. In the Classical and Roman eras (350BC - 150AD), Greek colonisation on the Taman Peninsula [7][8][9] and delta trading forts around Tanais paired with Scythian, Maeotian and Sarmatian populations in the hinterland (>3,6k studied burials) [10][11] to sustain 125k-132k inhabitants. After the post-Hunnic contraction, two medieval revivals carried the region to 150k under the Khazar Saltovo-Mayaki fortress and village network (900AD) [12][13] and 158k at the height of Golden Horde Azak (1340AD) [14][15]. Timur's sack of Azak in 1395AD reduced the population to 43k in 1400AD, but the rise of the 125 Don Cossack towns restored numbers to 120k by 1700AD [16][17][18]. Thereafter, peasant and Cossack ploughing of the black-earth steppes, port trade at Rostov-on-Don and Taganrog and eastern Donbas coal mining drove an extraordinary expansion to 550k in 1800AD, 1,4M in 1850AD and 3,95M by 1900AD [19][20][21][22][23][24][25].

Background

The Russian Pontic Steppe forms the land and river corridor between the North-Western Caucasus, the Sea of Azov (*Maeotis*), the Lower Don basin and the Caspian-Manych depression (Figure 1). Because rainfall and surface water vary sharply from the coast and floodplains to the dry eastern watersheds, settlement across this 165k km^2 area falls into four distinct regions. The first region comprises the Lower Don Delta, floodplain and Azov coast (18k km^2), encompassing Pochatkovsky (Porechny) Island (Rakushechny Yar), the Dead Donets (Mertvy Donets) and the main Don delta branches. Its principal settlements include Elizavetovskoye, Tanais, the ten Maeotian-Tanaitic delta forts (among them Kobyakovo, Nizhne-Gnilovskoye and Podazovskoye), Golden Horde Azak (Tana), Cossack Cherkassk and Novocherkassk and the modern port cities of Rostov-on-Don, Nakhichevan-on-Don, Taganrog, Azov and Yeysk [15][25][26]. The second region covers the Taman Peninsula, or Asian Bosporus (4,5k km^2, including the 1,2k km^2 ancient Taman archipelago and the Anapa/Gorgippia coast); surveyed by Ya. M. Paromov [7], V. D. Kuznetsov [27], U. Schlotzhauer and D. V. Zhuravlev [28], G. P. Garbuzov and A. A. Zavoikin [8][9] and A. V. Batasova [29], this deltaic peninsula hosted the Greek cities of Phanagoria, Hermonassa (later Tamatarkha/Tmutarakan), Kepoi, Patraeus and Gorgippia (Anapa), and from 1792 the Black Sea (Kuban) Cossack settlements [30].

The third region encompasses the Lower Don River Valley and Southern Don-Donets Corridor (77,5k km^2), comprising the main Lower Don valley and Tsimlyansk bend (37,5k km^2: Razdorskaya, Konstantinovsk, Semikarakorskoye, Sarkel/Belaya Vezha, Pravoberezhnoye Tsimlyanskoye and Cossack Razdory) together with the northern tributary valleys of the Seversky Donets, Chir, Kalitva, lower Khopyor and lower Medveditsa (40k km^2), which supported dense Late Bronze Age Srubnaya villages, Early Medieval Saltovo-Mayaki settlements, Don Cossack *stanitsy* and 19th-century coal-mining towns [6][12][22][31]. Finally, the fourth region consists of the Sal-Manych Watershed Steppe and Yergeni Slope (65k km^2), the dry inland steppes and seasonal grazing corridors linking the Lower Don to the Caspian Depression along the Western and Eastern Manych, Sal, Yegorlyk and Kalaus rivers (Manych Catacomb burial grounds, Khazar-period Bashanta I-II, Don Kalmyk pastures and 19th-century Salsk and northern Stavropol peasant districts) [3][4][13].

Methodology

Regional population estimates combine archaeological settlement surveys with historical registers and census returns. For the Mesolithic and Neolithic periods (Rakushechny Yar, Razdorskaya 2 and Kremennaya), excavated house floors (12-35 m^2 wattle-and-daub and *Viviparus* snail-shell platforms) and

radiocarbon dates [1][2][32][33] indicate riverine densities of 0,016-0,065 persons/km^2. For the Copper Age, Early Bronze Age (Yamnaya) and Middle Bronze Age (Catacomb), N. I. Shishlina's [3] strontium, carbon and nitrogen isotope studies of seasonal grazing routes are paired with A. V. Borisov and R. A. Mimokhod's [4] buried-soil analyses across the Sal-Manych and Yergeni steppes.

For the Late Bronze Age Srubnaya (Timber-Grave) and Kobyakovo cultures (2000-900BC), inventories of permanent river-terrace villages (*poseleniya*), excavated semi-subterranean timber-and-stone houses (5-20 dwellings of 80-250 m^2 per village) and household size estimates (8-12 persons per house) [5][6][34] are cross-checked against skeletal mortality series. For the Classical, Hellenistic and Roman/Sarmatian periods (600BC - 370AD), Ya. M. Paromov's [7] survey of >230 rural settlements on the Taman Peninsula (1,2k km^2), G. P. Garbuzov and A. A. Zavoikin's [8][9] agricultural land-use calculations, K. K. Marchenko, V. G. Zhitnikov and V. P. Kopylov's [26][35] estimates for Elizavetovskoye Gorodishche (44-55 ha, 3k-4,5k inhabitants) and M. A. Balabanova's [10] and E. F. Batieva's [11] skeletal mortality tables provide direct benchmarks.

For the medieval and modern periods (740-1900AD), G. E. Afanasyev's [12] Saltovo-Mayaki settlement study and A. N. Maslovsky and I. B. Papka's [14][15] >250-trench urban density study of Golden Horde Azak (54k inhabitants in 1340AD) are combined with 17th-century Russian *Posolsky* and *Razryadny Prikaz* rolls [16][17], 18th- and 19th-century imperial tax revisions (the ten *revizii* of 1719-1858) [19][20][21][22] and the First All-Russian Census of 1897 [23][24][25].[1]

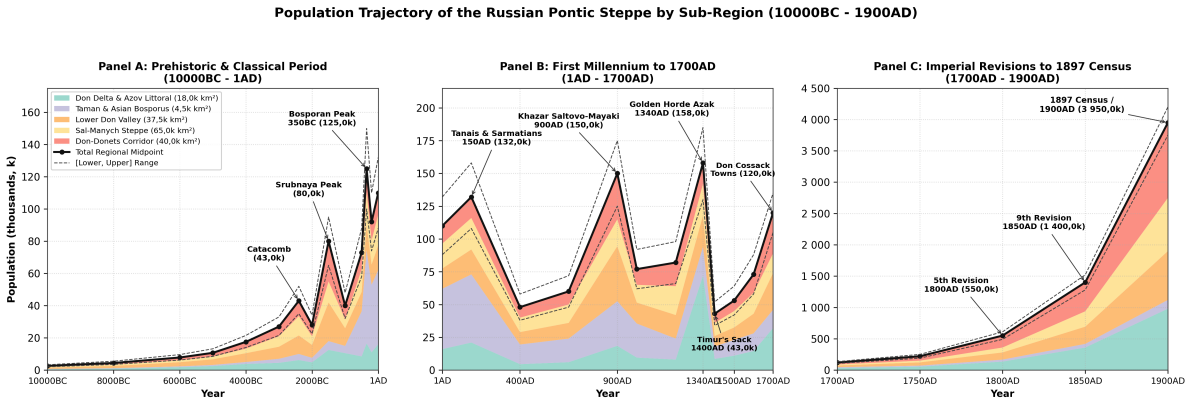


Figure 2. Panel A traces the four pre-modern population peaks across the Lower Don and Azov steppes: the Srubnaya river-terrace farming expansion (80k in 1500BC), the Classical and Roman Bosporan-Sarmatian era (125k-132k), the Khazar Saltovo-Mayaki network (150k in 900AD) and Golden Horde Azak (158k in 1340AD). Panel B shows the rapid expansion after 1700AD as Cossack and peasant farming, Azov grain ports and Donbas coal mining lifted the total from 120k to 3,95M by 1900AD.

Results

Early Fishing Communities and Bronze Age Herders

The Mesolithic and Early Neolithic

Studied by V. V. Tsybrij and A. V. Tsybrij [1], the Mesolithic sites of the Lower Don and Azov coast (Kremennaya II-III, Razdorskaya 2, Yuzhny Bugor, 10000-6200 cal BC) were flint-working camps and seasonal fishing stations on low floodplain terraces where inhabitants caught sturgeon (*Acipenseridae*), carp (*Cyprinidae*) and catfish (*Silurus glanis*) and hunted red deer and wild boar. Total population across

the 165k km^2 region rose gradually from 2,6k in 10000BC (0,016 persons/km^2) to 4,4k in 8000BC.

On Pochatkovsky (Porechny) Island near Stanitsa Razdorskaya in the Lower Don floodplain, the multi-layered settlement of Rakushechny Yar (excavated by T. D. Belanovskaya [36] and re-examined in 2008-2025 by E. V. Dolbunova, A. N. Mazurkevich, A. V. Tsybrij and V. V. Tsybrij [1][32][33]) contains 23 cultural layers reaching 5-6 m in depth. Radiocarbon and OSL dates place the earliest Neolithic shell-tempered pottery layers (Layers 23-15) between 6200/6000 and 5600 cal BC, making Rakushechny Yar one of the oldest pottery-producing villages in Europe [33]. Excavations uncovered artificial floors built from freshwater *Viviparus* snail shells (*rakovinnye platformy*), post-hole rows and clay-plastered wattle-and-daub (*turluk*) surfaces belonging to permanent riverine houses [32]. Individual floodplain villages housed 50-120 people (8-18 structures), raising Lower Don valley densities to 0,15-0,25 persons/km^2 and the regional population to 7,8k by 6000BC and 10,7k by 5000BC (Figure 2A).

The Copper Age and Early and Middle Bronze Age

During the Copper Age (5000-3300 cal BC), the Lower Don (Nizhnedonskaya), Sredny Stog/Novodanilovka and Konstantinovka traditions established river-terrace farming, herding and fishing villages (1-2,5 ha settlements with 60-150 residents each at Razdorskoye I and Samsonovskoye) alongside flat and low-mound cemeteries (Karatayevo, Mukhin). In an analysis of Layer 5 at Rakushechny Yar (a 55 cm-thick *Viviparus* shell midden dated to 4600-4300 cal BC), A. M. Skorobogatov, E. V. Dolbunova and A. N. Mazurkevich [2] showed that collared and comb-impressed Nizhnedonskaya pots make up 6,6% of the ceramic assemblage alongside local Late Neolithic vessels, consistent with a regional population of 17,5k by 4000BC.

In her study *Reconstruction of the Bronze Age of the Caspian Steppes* [3], N. I. Shishlina showed that the transition to the Yamnaya (Pit-Grave, 3300-2600 cal BC) and Katakombnaya (Catacomb, 2600-2000 cal BC) periods introduced four-season mobile herding with heavy four-wheeled ox-wagons. Whereas Yamnaya herders moved along short river-valley loops (10-25 km), population growth during the Manych Catacomb phase (2500-2100 cal BC) led to longer north-south seasonal migrations (150-300 km routes linking the Lower Don and Manych lakes to the Stavropol Uplands and Yergeni Hills). Field surveys and buried-soil studies by A. V. Borisov and R. A. Mimokhod [4] record burial mound densities of 0,4-1,2 kurgans/km^2 along river terraces and 0,1-0,3 kurgans/km^2 on open watersheds, with 60-75% of all Bronze Age burials belonging to the Catacomb period. Heavy grazing contributed to alkaline soil (*solonetz*) formation before the 4,2 ka BP (2200-2000 cal BC) dry phase, which emptied the open Sal-Manych watersheds during the post-Catacomb Lola period [4]. Regional population rose from 27k in 3000BC to 43k during the Catacomb period (2400BC), before falling to 28k around 2000BC (Figure 3A).

Late Bronze Age Farming and Herding Villages

With wetter conditions returning after 1850/1750 cal BC, the Srubnaya (Timber-Grave) Culture (Berezhnovka-Mayovka group) produced a dense network of permanent farming and herding villages across the Lower Don, Seversky Donets, Chir, Khopyor and Azov coastal valleys. As summarised by E. S. Sharafutdinova [34] and V. V. Potapov [5][6], field surveys in Rostov Oblast and the southern Don basin have recorded more than 1,2k permanent Srubnaya settlements (*poseleniya*) along first and second river terraces and ravine springs, spaced 1,5-3,5 km apart.

Unlike temporary Yamnaya or Catacomb camps (*stoyanki*), Srubnaya villages (0,8-4 ha) consisted of 5 to 20 large semi-subterranean timber-framed and limestone-lined houses (*poluzemlyanki*), each measuring 80 to 250 m^2. Animal bone assemblages studied by V. V. Potapov [6] and V. R. Erlikh et al. [37] show settled herds dominated by cattle (50-65% cattle, alongside horses, sheep, goats and pigs). At 10-14 people per extended household, an average Srubnaya village of 8-12 occupied houses held 80-150 residents.

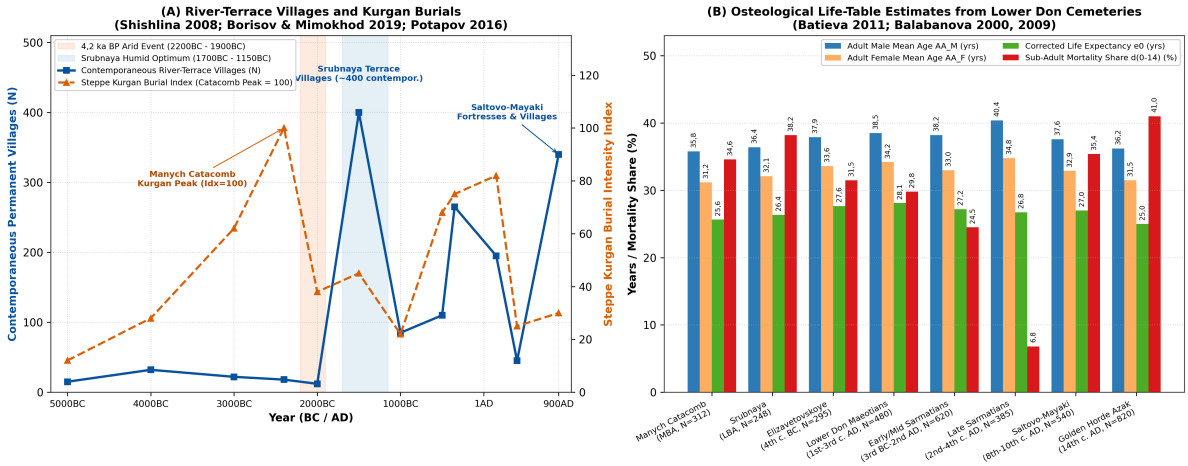


Figure 3. Panel A compares river-terrace village counts and open-steppe kurgan densities across the Bronze Age and Early Iron Age, showing the shift from Manych Catacomb mobile herding to over 1,2k permanent Srubnaya villages around 1500BC and the subsequent dry-steppe pullback. Panel B contrasts child mortality and adult ages at death across eight excavated burial series, highlighting the ritual absence of young children in Late Sarmatian mounds and the high youth mortality in 14th-century Azak.

Assuming that roughly 30% of the >1,2k recorded terrace settlements were occupied at the same time within any 100-year phase (350-450 contemporaneous villages), the Srubnaya population of the Lower Don and Azov steppes reached 80k (65k-95k) around 1500BC (0,485 persons/km^2 overall; 1,5-2,8 persons/km^2 in the river valleys). During the Final Bronze Age (1200-900 cal BC), drier conditions reduced open-steppe village numbers and house sizes [5], while the stone-built Kobyakovo Culture [34] concentrated 15-25 fishing and bronze-working villages in the Don Delta (Kobyakovo, Safyanovo, Nizhne-Gnilovskoye), sustaining a regional population of 40k around 1000BC.

Greek Colonies, Tanais and Steppe Nomads

Cities and Rural Settlements of the Taman Peninsula and Gorgippia

On the Taman Peninsula (*Asian Bosporus*, an ancient archipelago of three main islands covering ~ 1,2k km^2), Ya. M. Paromov [7] carried out a full field and aerial survey that mapped >230 Greek and Sindian-Maeotian rural settlements (*chora*), fortified villages (such as Golubitskaya 2 at 7,2 ha, excavated by U. Schlotzhauer and D. V. Zhuravlev [28]), rectangular field systems and farmsteads (Figure 4A).

As reconstructed by G. P. Garbuzov, A. A. Zavoikin, A. A. Maslennikov and S. L. Smekalov [8][9] and A. V. Batasova [29], Greek settlement on Taman developed in two main stages. During the Late Archaic settlement phase (580-480BC), following the foundation of Hermonassa, Kepoi, Patraeus, Phanagoria (c. 542BC, founded by Teos) and inland settlements at Beregovoy 4, Golubitskaya 2 and Akhtanizovskaya 4, rural farmsteads spread across the peninsula, growing from 12-15 early sites to 70-93 contemporaneous rural settlements by 500-480BC [8][29]. During the Classical and Early Hellenistic Spartocid period (400-250BC), after Sindika was incorporated into the Bosporan Kingdom, 180-215 rural villages and estates were occupied at the same time on the Taman archipelago [7][9]. Phanagoria covered 65 ha (~ 43 ha on land plus 22 ha now submerged in Taman Bay), housing 10k-15k urban residents [27]. Together with Hermonassa (25-30 ha, 4,5k-6,5k), Kepoi (20-25 ha, 3k-4,5k), Patraeus (12-15 ha, 1,5k-2,5k), Gorgippia (modern Anapa, 40 ha, 6k-9k) and 20k-28k rural *chora* inhabitants, the Asian Bosporus supported 45k-65k people (58k midpoint in 350BC), corresponding to 37,5-54,2 persons/km^2 on Taman.

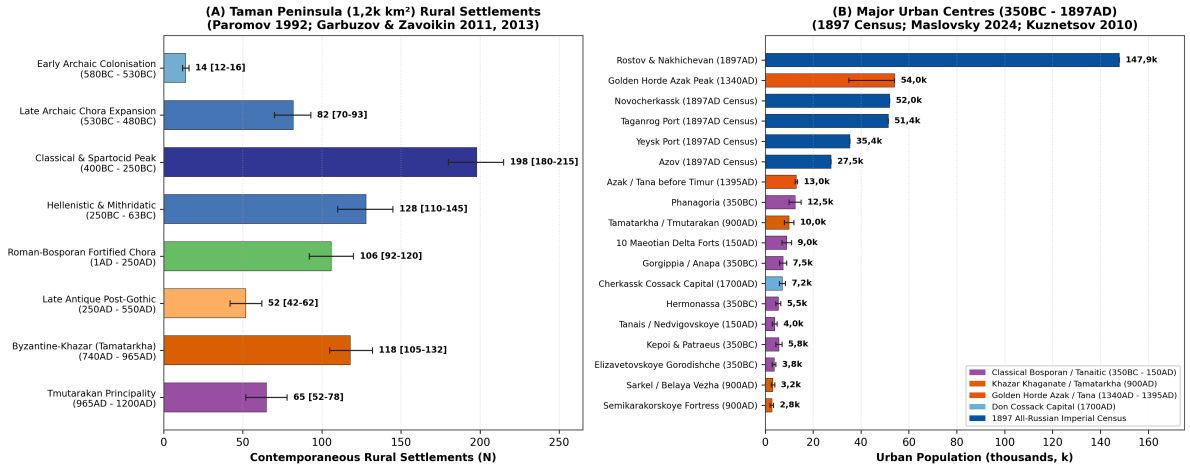


Figure 4. Panel A tracks the number of occupied rural villages on the Taman Peninsula across the Archaic, Spartocid Hellenistic (180-215 sites in 350BC), Roman and Khazar periods. Panel B compares the urban populations of Phanagoria, Tanais, Sarkel, Golden Horde Azak (54k in 1340AD before falling to 13k on the eve of Timur's 1395 sack) and the 19th-century port cities of Rostov-on-Don, Novocherkassk and Taganrog.

Lower Don Delta Trading Towns and Scythian-Sarmatian Nomads

In the Lower Don Delta, Elizavetovskoye Gorodishche (5th - early 3rd c. BC), excavated by K. K. Marchenko, V. G. Zhitnikov and V. P. Kopylov [26][35], covered 44-55 ha with a 12 ha inner citadel enclosed by two 4th-century BC ditch-and-rampart lines that housed a Bosporan Greek merchant quarter and later colony, supporting 3k-4,5k permanent inhabitants. Terracotta and pottery studies by N. I. Suvorova [38] show direct population continuity between the abandonment of Elizavetovskoye (c. 275-270BC) and the founding of Tanais (Nedvigovskoye Gorodishche) on the right bank of the Dead Donets (16-20 ha, 3k-5k residents). Between the 1st c. BC and 3rd c. AD, Bosporan rulers settled Kuban Maeotian farming and military communities in a ring of ten fortified delta towns (*gorodishcha*: Kobyakovo, Nizhne-Gnilovskoye, Podazovskoye, Krepkoye, Sukho-Chaltyrskoye, Temernitskoye, Khopry, Safyanovo, Lebyazhye and Rogozy), which held another 7k-11k settled inhabitants [39] until the Gothic-Borani raids of 244-254AD.

Across the surrounding Don-Manych-Azov steppes, E. F. Batieva [11] and M. A. Balabanova [10] examined >1,5k Scythian, Maeotian and Sarmatian skeletons, finding adult male mean ages at death of 37,9-40,4 years, adult female ages at death of 33-34,8 years and corrected life expectancy at birth of 25,5-29,2 years (Table 2). Using A. S. Skripkin's [40] Kalmyk steppe pasture comparison (0,35-0,65 persons/km^2), the nomadic Sarmatian tribes (Siraces, Aorsi and Alans) numbered 40k-65k in the 1st-2nd c. AD, giving a combined regional population of 132k (108k-158k) in 150AD.

Khazar Fortresses, Golden Horde Azak and the Early Don Cossacks

The Khazar Khaganate and the Saltovo-Mayaki Culture

After population declined during the Hunnic and post-Hunnic centuries (48k in 400AD; 60k under Kubrat's Old Great Bulgaria in 650AD), the Khazar peace fostered widespread village and town settlement (the Saltovo-Mayaki Culture) along the Lower Don, the Tsimlyansk bend, the Manych valley and the Don-Donets borderlands [12][31][41]. Along the Lower Don, the network was anchored by the brick fortress of Sarkel (Levoberezhnoye Tsimlyanskoye, built c. 834-837AD; 3,7 ha citadel plus >15 ha outer town, 2,5k-4k inhabitants) [31], the white-stone fortress of Pravoberezhnoye Tsimlyanskoye (1,5k-2,5k)

and the mudbrick fortress of Semikarakorskoye (3,8 ha citadel plus >30 ha settlement, 2k-3,5k) [41], surrounded by >120 open farming and wine-growing villages (*selishcha*). In the southern Sal-Manych/Western Kalmyk valley, M. A. Ochir-Goryaeva [13] excavated the twin Khazar-period fortified settlements of Bashanta I and Bashanta II, confirming permanent farming and herding villages along the Yergeni-Manych streams.

On the Taman Peninsula, Ya. M. Paromov [7] mapped >110 Khazar-period (8th-10th c. AD) rural villages (with semi-dugouts and stone-foundation houses similar to Artezian [42]) around the port city of Tamatarkha (Hermonassa, ~ 35 ha, 8k-12k residents), supporting 34k inhabitants on the Asian Bosporus. Combined with G. E. Afanasyev's [12] Don-Donets Saltovo-Mayaki valley settlements (5-8,5 persons/km^2), total regional population reached 150k (125k-175k) in 900AD, before Sviatoslav's 965AD campaign and Pecheneg-Cuman nomadism reduced the regional total to 77k-82k in 1000-1200AD.

Golden Horde Azak and the Rise of the Don Cossack Host

Founded around 1269-1270AD under Khan Mengu-Timur at the mouth of the Don (modern Azov), Azak (together with the Venetian and Genoese merchant quarters of Tana) served as the main western port of the Golden Horde. Drawing on rescue excavations across >250 trenches in modern Azov, A. N. Maslovsky [14][43] and A. N. Maslovsky and I. B. Papka [15] reconstructed the growth and contraction of the city (Figure 4B). During the early settlement phase around 1290AD, the urban population numbered 3k-3,5k inhabitants. At the height of urban expansion under Uzbek and Janibek in 1340AD, Azak covered >400 ha with paved streets (3-8 m wide) and closely packed wattle-and-daub and mudbrick courtyard houses heated by underfloor *kan* flues and *tandyr* ovens, reaching an estimated 54k urban residents (35k-54k range) [15]. During the civil wars of the *Velikaya Zamyatnya* (1359-1380AD) and the reign of Tokhtamysh (1381-1395AD), political conflict and recurring plague epidemics reduced the inhabited area to a quarter of its former size and eliminated the outer artisan dugouts (*zemlyanki*) [14], leaving 12,5k-13,5k inhabitants on the eve of Timur's sack in September 1395AD [15].

Together with 18k rural delta farmers and 86k Kipchak-Tatar nomads in the Don-Manych-Azov steppes (part of a wider Azov-Kuban nomadic population estimated at up to 250k by Maslovsky and Papka [15]), the Lower Don and Azov steppes reached 158k in 1340AD before dropping to 43k in 1400AD.

After the Ottoman conquest of Azak in 1475AD (leaving an Ottoman fortress town of 3,5k-6k residents) and the formation of the Lesser Nogai Horde (*Kazyev Ulus*, 25k-40k nomads), free Don Cossack communities (*Voysko Donskoye*) expanded from 4-6 early river-island strongholds (Razdory, Donskoy/Donetsky [18] and Cherkassk, founded in 1570) to 30-40 fortified towns (*gorodki*) by 1637 (22k-28k people) and 125 *gorodki* along the Don (56), Seversky Donets (11), Khopyor (23), Medveditsa (19) and Buzuluk (16) by 1695-1700AD [16]. As shown by S. Ya. Sushchiy [17], early family formation on the Lower Don (initially involving captive Turkish and Tatar women) was reinforced after 1650 by a steady influx of married Russian peasant families into the Upper Don towns. By 1700AD, following Peter I's 1696 capture and garrisoning of Azov and Taganrog, the Don Cossack Host numbered 65k-80k people, bringing the regional population to 120k (105k-135k).

Cossack Stanitsas, Peasant Colonisation and Azov Port Cities

The Don Land, Taganrog, Rostov and the Black Sea Cossacks

During the early 18th century, the suppression of the Bulavin Rebellion (1707-1708) and the Temporary Transfer of Azov back to Ottoman rule (1711-1739) slowed growth along the lower river, but Russian and Ukrainian peasant migration onto the estates of Don Cossack elders (*starshina*) soon accelerated [19][21]. By 1750AD, follow-up military rolls and the Second Imperial Revision recorded ~220k inhabitants across the study region, including 55k in the Don Delta and Taganrog coast (following the establishment of the

Temernitskaya custom house in 1749), 122k across the Lower Don and Don-Donets *stanitsy* and peasant *slobody*, 27k Don Kalmyk and Nogai herders in the Sal-Manych steppe and 16k residents on the Taman Peninsula [19][20].

Following the Russo-Turkish War of 1768-1774 (Treaty of Küçük Kaynarca) and the annexation of the Azov-Kuban coast (1783), imperial settlement transformed the coastal and southern steppes. Taganrog was rebuilt as a naval and commercial port in 1769, the Fortress of St. Dimitry of Rostov (1761) grew into Rostov-on-Don and in 1779 Catherine II resettled >12k Crimean Armenians on the adjacent right bank at Nakhichevan-on-Don and five surrounding villages (Chaltyr, Krym, Sultan-Saly, Nesvetay and Bolshiye Saly) [20][25]. Between 1777 and 1783, the construction of the Azov-Mozdok fortified line encouraged state peasant settlement along the Yegorlyk and Manych valleys, and in 1792-1793 the Black Sea Cossack Host (*Chernomorskoye kazachye voysko*, ~25k former Zaporozhian Cossacks) landed on Taman and settled 40 *kureni* across the Taman Peninsula and right-bank Kuban steppes [30][44]. As documented by A. P. Pronshtein [19] and V. M. Kabuzan [20] in the Fifth Imperial Revision (1795/1800AD), the registering of Ukrainian and Russian peasants attached to Don officer estates alongside the Cossack *stanitsy* raised the population of the Russian Pontic Steppe to 550k (490k-610k) by 1800AD (Figure 2B).

Peasant In-Migration, Grain Ports and Railways

In the first half of the 19th century, the transfer of the Don Host capital from flood-prone Starocherkassk to Novocherkassk (founded by Ataman M. I. Platov in 1805), the opening of the port of Yeysk on the Azov coast (1848) and the expansion of wheat exports through Taganrog and Rostov-on-Don drove rapid population growth. By the Ninth Revision of 1850, the study region contained 1,4M inhabitants, including 360k in the Don Delta and Azov port districts, 737k along the Lower Don and Don-Donets valleys, 245k in the Sal-Manych and northern Stavropol steppes and 58k on the Taman Peninsula [20][21].

After the Emancipation Reform of 1861, the end of the Caucasian War (1864) and the 1868 law permitting non-Cossack citizens (*inogorodniye*) to buy and lease property across the Don and Kuban territories, hundreds of thousands of land-short peasants from the Ukrainian and Central Russian Black-Earth governorates migrated into the region [22][30][45]. As M. A. Kolomeytseva [22] demonstrated, high rural birth rates (48-55 per 1k) and peasant immigration coincided with the opening of the Grushevsky (Aleksandrovsk-Grushevsky/Shakhty) and Sulin anthracite coalfields, metallurgical plants at Taganrog and Sulin and the Voronezh-Rostov and Vladikavkaz railways.

The First All-Russian Census of 28 January 1897 [23][24] provides a comprehensive baseline for the end of the 19th century. The Province of the Don Cossack Host (*Oblast Voyska Donskogo*) recorded 2,56M inhabitants across 164,5k km^2 (12,4% urban), of whom ~2,15M lived within the Lower Don, Don Delta/Taganrog, Donetsk, Sal-Manych and southern Ust-Medveditsa/Khopyor districts inside the study boundary (Rostov-on-Don 119,5k plus Nakhichevan-on-Don 28,4k = 147,9k combined town; Novocherkassk 52k; Taganrog 51,4k; Azov 27,5k; Aleksandrovsk-Grushevsky 16,5k) [23][25]. Within the southern portion of the 165k km^2 study area, the Temryuksky (Taman), Yeysky (Yeysk city 35,4k) and northern Kavkazsky departments of Kuban Oblast [24][30] together with the Manych/Medvezhensky steppe districts of northern Stavropol Governorate accounted for a further 1,55M-1,65M residents in 1897. Projecting forward at the observed 1,8% annual growth rate over the three years from January 1897 to 1900AD gives a regional population of 3,95M (3,75M-4,2M, or 23,94 persons/km^2) by 1900AD (Table 3).

Settlement / Site	Region	Chronology	Core / Total Area (ha)	Population Estimate	Primary Literature
Rakushechny Yar	Lower Don (Pochatkovsky Isl.)	6200-4300 cal BC	1,2 (core) / 2,5	80-150	Tsybrij et al. 2018; Skorobogatov 2025
Kobyakovo (Final Bronze)	Lower Don Delta	1200-900 cal BC	3,5 (core) / 8	300-500	Sharafutdinova 1980; Potapov 2013
Phanagoria	Taman Peninsula	542BC-10th c. AD	43 (land) / 65	10k-15k (350BC)	Kuznetsov 2010; Garbuzov 2013
Gorgippia (Anapa)	Asian Bosporus Coast	5th c. BC-240AD	35 / 40	6k-9k (350BC)	Paromov 1992; Garbuzov 2013
Hermonassa / Tamatarkha	Taman Peninsula	6th c. BC-14th c. AD	25 / 35	4,5k-6,5k / 8k-12k	Garbuzov 2011; Batasova 2015
Kepoi & Golubitskaya 2	Taman Peninsula	6th c. BC-4th c. AD	20-25 / 7,2-9	3k-4,5k / 0,6k-1k	Schlotzhauer & Zhuravlev 2014
Elizavetovskoye Gorodishche	Lower Don Delta	500-270BC	12 (cit.) / 44-55	3k-4,5k (4th c. BC)	Marchenko et al. 2000; Kopylov 2016
Tanais (Nedvigovskoye)	Dead Donets (Don Delta)	275BC-450AD	14 / 20	3k-5k (150AD)	Suvorova 2025; Vdovchenkov 2019
10 Maeotian Delta Forts	Lower Don Delta	1st c. BC-250AD	0,5-4 / 10-35 ea.	7k-11k (combined)	Vdovchenkov 2019
Sarkel (Belaya Vezha)	Lower Don (Tsimlyansk)	834-1117AD	3,7 (fort) / 18	2,5k-4k (900AD)	Artamonov 1958; Afanasyev 1993
Semikarakorskoye & Bashanta	Lower Don & Manych	8th-10th c. AD	3,8 (fort) / 32	2k-3,5k (900AD)	Flyorov 2011; Ochir-Goryaeva 2020
Azak / Tana (Azov)	Lower Don Delta	1270-1395AD	120 / >400	54k (1340) / 13k (1395)	Maslovsky 2022; Maslovsky & Papka 2024
Cherkassk (Starocherkassk)	Lower Don	1570-1805AD	35 / 60	6k-8,5k (1700) / 15k (1800)	Mininkov 1998; Pronshtein 1961
Rostov & Nakhichevan-on-Don	Lower Don Delta	1749/ 1779-1900AD	680 / 1,25k	147,9k (1897 Census)	Troynitsky 1905; Volvenko 2023
Novocherkassk (Host Capital)	Lower Don Basin	1805-1900AD	320 / 550	52k (1897 Census)	Troynitsky 1905; Agafonov 2017
Taganrog & Yeysk (Azov Ports)	Taganrog Gulf / Azov Coast	1698/ 1848-1900AD	410 / 720	51,4k & 35,4k (1897 Census)	Troynitsky 1905; Volvenko 2023

Table 1. Excavated core and total areas (in hectares) and estimated populations for principal towns and fortresses of the Lower Don, Taman Peninsula and Azov coast from Neolithic Rakushechny Yar to the 1897 Census.

Limitations

Three practical constraints affect population estimates in the Russian Pontic Steppe. First, both the Taman Bay (where rising sea levels after the Classical period submerged ~ 22 ha of Phanagoria and coastal parts of Kepoi and Patraeus) [8][27] and the advancing Don Delta (where 5-7 m of river silt cover the prehistoric floodplain) conceal low-lying Mesolithic, Neolithic and Archaic shoreline camps. Second, as shown by M. A. Balabanova [10] and E. F. Batieva [11], Late Sarmatian (2nd-4th c. AD) burial mounds rarely include infants and young children (only 6,8% sub-adults) and contain more than twice as many adult men as women; standard life-table corrections (life expectancy at birth of 25,5-28 years) are therefore required instead of unadjusted skeletal averages. Third, nomadic groups (Manych Catacomb herders, Scythians, Sarmatians, Cumans, Golden Horde *uluses*, Lesser Nogais and Don Kalmyks) moved seasonally across the Sal-Manych, Kuban and Stavropol steppes [3][15], while early 18th-century Cossack and runaway peasant settlements frequently evaded imperial tax enumerators prior to the Fifth Revision of 1795 [19][21].

Conclusion

Before the 18th century, the Russian Pontic Steppe experienced four pre-modern waves of dense settlement: the Late Bronze Age Srubnaya river-terrace farming expansion (80k around 1500BC), the Classical and Roman Bosporan and Sarmatian period (125k-132k in 350BC - 150AD), the Khazar Saltovo-Mayaki fortress and village network (150k around 900AD) and the Golden Horde urban expansion centred on Azak/Tana (158k in 1340AD). Following Timur's destruction of Azak in 1395AD (43k in 1400AD) and the growth of the 125 Don Cossack towns to 120k by 1700AD, 18th- and 19th-century Russian, Ukrainian, Armenian and German colonisation of the black-earth steppes, Azov port trade and eastern Donbas coal mining raised the regional population more than thirtyfold to 550k in 1800AD, 1,4M in 1850AD and 3,95M (23,94 persons/km^2) by 1900AD.

Appendix

Burial Mortality Estimates

Archaeological Period	Sample (N)	Child Share (0-14)	Male AA (yrs)	Female AA (yrs)	Corrected e_0 (yrs)	Sources
Manych Catacomb (MBA)	312	34,6%	35,8	31,2	24,5-26,8	Batieva 2011
Srubnaya (LBA, Lower Don)	248	38,2%	36,4	32,1	25,2-27,5	Balabanova 2009; Batieva 2011
Elizavetovskoye (4th c. BC)	295	31,5%	37,9	33,6	26,8-28,5	Batieva 2011; Marchenko et al. 2000
Lower Don Maeotians (1st-3rd c. AD)	480	29,8%	38,5	34,2	27-29,2	Batieva 2011
Early/Middle Sarmatians (3rd c. BC-2nd c. AD)	620	24,5% (kurgan bias)	38,2	33	26-28,4	Balabanova 2009; Batieva 2011
Late Sarmatians (mid-2nd-4th c. AD)	385	6,8% (burial filter)	40,4	34,8	25,5-28	Balabanova 2009; Batieva 2011

Saltovo-Mayaki (Sarkel & Lower Don)	540	35,4%	37,6	32,9	25,8-28,2	Afanasyev 1993; Batieva 2011
Golden Horde Azak (14th c. AD)	820	41% (plague/ sack)	36,2	31,5	23,5-26,5	Batieva 2011; Maslovsky 2024

Table 2. Sample sizes, child burial shares, mean adult ages at death (AA) and life-table corrected life expectancy at birth (e_0) across 3,7k skeletons excavated in the Lower Don and Azov steppes.

Population Estimates

Year	Don Delta (18k km²)	Taman (4,5k km²)	Lower Don (37,5k km²)	Sal-Manych (65k km²)	Don-Donets (40k km²)	Total Population [Range]	Density (p/km²)
10000BC	0,5k	0,2k	0,8k	0,5k	0,6k	2,6k [2k-3,3k]	0,016
8000BC	0,9k	0,3k	1,4k	0,8k	1k	4,4k [3,5k-5,5k]	0,027
6000BC	1,8k	0,5k	2,6k	1,2k	1,7k	7,8k [6,2k-9,6k]	0,047
5000BC	2,5k	0,7k	3,5k	1,6k	2,4k	10,7k [8,5k-13,2k]	0,065
4000BC	3,8k	1,2k	5,5k	2,8k	4,2k	17,5k [14k-21,5k]	0,106
3000BC	4,5k	2,5k	7,5k	6,5k	6k	27k [21,5k-33k]	0,164
2400BC	6,2k	3,8k	11,5k	12k	9,5k	43k [35k-52k]	0,261
2000BC	4,8k	2,8k	8k	5,4k	7k	28k [22,5k-34k]	0,17
1500BC	12,5k	5,5k	24k	13k	25k	80k [65k-95k]	0,485
1000BC	10,5k	4,5k	11k	5,5k	8,5k	40k [32k-48k]	0,242
500BC	8,5k	28k	12k	14k	10,5k	73k [58k-88k]	0,442
350BC	16,5k	58k	16k	20k	14,5k	125k [100k-150k]	0,758
200BC	11k	42k	12,5k	15k	11,5k	92k [74k-110k]	0,558
1AD	16k	46k	15,5k	19k	13,5k	110k [88k-132k]	0,667
150AD	21k	52k	19k	24k	16k	132k [108k-158k]	0,8
400AD	4,5k	15k	9,5k	11k	8k	48k [38k-58k]	0,291
650AD	6k	18k	12k	14k	10k	60k [48k-72k]	0,364
900AD	18,5k	34k	42k	20,5k	35k	150k [125k-175k]	0,909
1000AD	9,5k	26k	16k	13,5k	12k	77k [62k-92k]	0,467
1200AD	8k	16k	18k	22k	18k	82k [66k-98k]	0,497
1340AD	72k	22k	24k	24k	16k	158k [130k-185k]	0,958
1400AD	8,5k	9,5k	8k	10k	7k	43k [34k-52k]	0,261
1500AD	11k	12k	9,5k	12,5k	8k	53k [42k-64k]	0,321
1600AD	14,5k	13,5k	15k	16k	14k	73k [58k-88k]	0,442
1700AD	32k	14k	28k	15k	31k	120k [105k-135k]	0,727
1750AD	55k	16k	52k	27k	70k	220k [190k-250k]	1,33
1800AD	135k	32k	115k	78k	190k	550k [490k-610k]	3,33
1850AD	360k	58k	275k	245k	462k	1,4M [1,28M-1,52M]	8,48

1900AD	980k	140k	780k	850k	1,2M	3,95M [3,75M-4,2M]	23,94

Table 3. Population of the Lower Don Delta, Taman Peninsula, Lower Don valley, Sal-Manych steppe and Don-Donets corridor at 29 benchmark dates, with combined regional totals, uncertainty ranges and overall density across the 165k km^2 study area.

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Footnotes

[1] All archived full-text academic PDFs cited in this review are stored locally in histmap/1.data_raw/population_regional/POLARIS/pontic_steppe_and_lower_volga/papers/.